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RESEARCH FEATURE

Therac-25 Accidents: We Keep on Learning From Them

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A medical physicist describes events around an accident in which a radiation therapy machine killed two patients due to software bugs, a poor quality user interface, an obscure error message, and the machine's failure to shut itself down permanently after noting that something was very wrong. The article ends with some lessons.

Radiation therapy (RT) is, along with surgery and chemotherapy, an effective technique to treat cancer. The treatment takes place in a computer-controlled linear accelerator (linac) that rotates its gantry around the patient, targeting the tumor with high-energy particles beams, coming from multiple directions (Figure 1). Due to severe occupational risks involved, modern RT systems and processes are strictly regulated by safety standards.[1,2]

However, back in the 1980s, things used to be different. Software development was a small discipline rather than a mass industry, and the RT safety legislation was less rigorous. This was the time when Therac-25, the first computer-controlled linac,[4] promoted as the "Rolls Royce of accelerators," [5] was manufactured by the Atomic Energy of Canada Limited (AECL). Unfortunately, in 1986–1987, the same Therac-25 entered the history as being responsible for six major radiation overexposure accidents.[6,7,8] The events shocked the world and set back the linac industry for 25 years.

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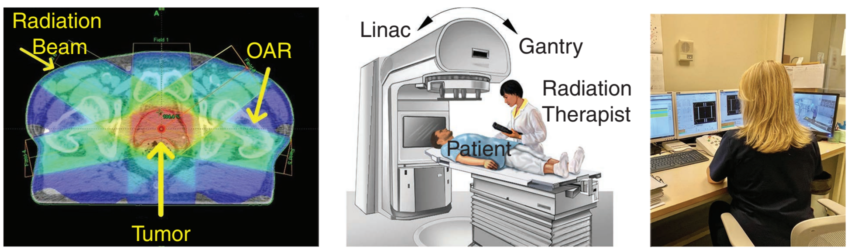
Because the root causes resided in immature, flawed software engineering practices, we have been using the Therac-25 accidents for many years already to sensitize and motivate software testing novices.[9] In 2019, we initiated VU-BugZoo, an ambitious educational project that aimed among other things, to build a physical replica of the Therac-25 linac to demonstrate a particular type of accidents, known as "Malfunction-54." Very soon, however, we discovered that an accurate replication is not trivial. To start with, we could not find any picture of the machine. Also, we did not know how the user console screen looked like. Finally, we did not understand why the accidents happened after all, if in the control room the operator could see a box showing the state of the machine. If you, the reader, already know the answers to all of these questions, then you can stop reading this article as it will hardly bring anything new. However, if you do not, please keep on reading.

After a long quest, in March 2021 we reached Dr. Fritz Hager, retired medical physicist who closely witnessed two Malfunction-54 accidents. This was the same Fritz Hager praised by Leveson for his diligence, because "without his efforts the understanding of the software problems might have been delayed even further." [6] We organized a few online meetings where teachers and students interviewed Dr. Hager. Were these meetings useful? We think yes. First, because they answered our technical questions and even revealed new details. Second, because if we paraphrase Eli Wiesel,[10] by listening to a witness, our students became witnesses themselves, realizing how the harm of suboptimal systems engineering extends toward innocent users, also called "second victims," [11] creating psychological pressure and long-lasting emotions, like despair, guilt, and regret. As a result, our students received an extra motivation to thoroughly test their systems in the future and investigate accidents when they happen against all odds.

Hence this article, because we believe that leaving things untold would be wrong, and even unfair with respect to other computing educators and professionals. After a short introduction to the field of RT, an uncharted territory for most computer professionals, we will share (with permission) the most important fragments of our interviews, relevant to readers interested in an accurate historical reconstruction of the Therac-25 accidents. We will end up with a few lessons useful for educators and computing professionals who are considering a career in safety-critical industry.

RT IN A NUTSHELL

To thoroughly understand the events, let us first explain how RT works. The treatment starts with a radiation oncologist prescribing the needed clinical parameters, such as the radiation dose that the tumor must receive, usually divided in daily fractions. A typical daily dose is around 200 rad, whereas a dose of 1,000 rad can be fatal for a human being. In an RT facility, a team of radiation therapists and medical physicists designs a treatment plan, specifying the kind of particles to use (electrons or photons, also called X-rays), their energy and orientation, with the aim to obtain a maximal radiation dose in the tumor, while protecting the surrounding healthy tissue and organs at risk (OAR) [Figure 1(a)]. Next, a team of radiographers prepares the linac, positions the patient on the treatment table [Figure 1(b)] in the treatment room and proceeds to the control room, to remotely control the linac and treat the patient according to the plan [Figure 1(c)]. A simple push on the *Beam ON* button at the console will trigger a powerful treatment delivery process that starts with electrons being generated in an electron gun (Figure 2). From there, electrons are accelerated to high energies (usually 5 MeV to 25 MeV), and bent downwards by 270° deflecting magnets, to reach a vertical position.[12] A few extra components need to be present in the beam path. For example, in a treatment with photons, a metallic target and a much higher current density electron beam are needed to extract the X-rays. Other components are beam-shaping and attenuating devices, such as collimators, trimmers, and filters. Another essential component always present in the beam path is a dose monitor, continuously measuring the beam intensity that will reach the patient. Dual-beam linacs are more complex machines that allow treatment with both electrons and photons. They usually require a motor-driven turntable to bring the right accessories at the right moment in the beam path. Eventually, the correctly shaped beam of high-energy ionizing particles exits the gantry, heading with high precision toward the patient and their tumor(s). The patient's well-being is monitored via audio and video communication. Any time when the situation is likely



(a) (b) (c)

FIGURE 1. (a) The principle of RT. Adapted with permission from Kane.[3] (b) A patient about to receive RT in the treatment delivery room. Credits to National Cancer Institute (NCI). Adapted with permission. (c) A radiation therapist at work in the control room. Credits to Anastasia Sarchosoglou.

to become unsafe, such as a radiation overdose, the machine must pause with an error message, leaving the therapist at the console to decide whether to resume or to abort the treatment.

THERAC-25 ACCIDENTS THROUGH THE EYES OF A WITNESS

PROFESSOR AND STUDENTS: Fritz, thank you for accepting our invitation. Maybe it is good to start by describing your role and duties at the Tyler East Texas Cancer Center (ETCC) facility.

FRITZ HAGER: I was a medical physicist. My role was to ensure that the linacs were in good condition and that the treatment was delivered safely and according to the oncologist's medical prescription.

PROFESSOR AND STUDENTS: Our first question is whether the images found on the Internet are showing the real Therac-25.

HAGER: No. All of these pictures are showing a newer type of linac. The reason you cannot find any visual information about the Therac-25 is because immediately after the accidents, AECL retrieved all of the user manuals related to this particular machine. I only have a newspaper cut from an old *People's Weekly Magazine*[13] showing the real Therac-25 [Figure 3(a)].

PROFESSOR AND STUDENTS: In [Figure 3(b)], we see another photo from your private collection, showing the flash of light created by the flawed machine that you recorded during the reproduction of the accidents. It gives an idea of the enormous radiation dose that was received by the unfortunate patients. But about this later.

PROFESSOR AND STUDENTS: The next question is related to the Therac-25 construction. We are all familiar with the top view of the gantry head as described in Leveson's paper. Can you help us reconstruct a cross-section view?

HAGER: The difference between the two modes E (electrons) and X (X-rays) only occurred in the gantry head, after the electron beam exited the 270° bending magnet section. Then you either had in the electron beam path two pairs of scanning magnets to spread the electron beam in E-mode, or a target-primary

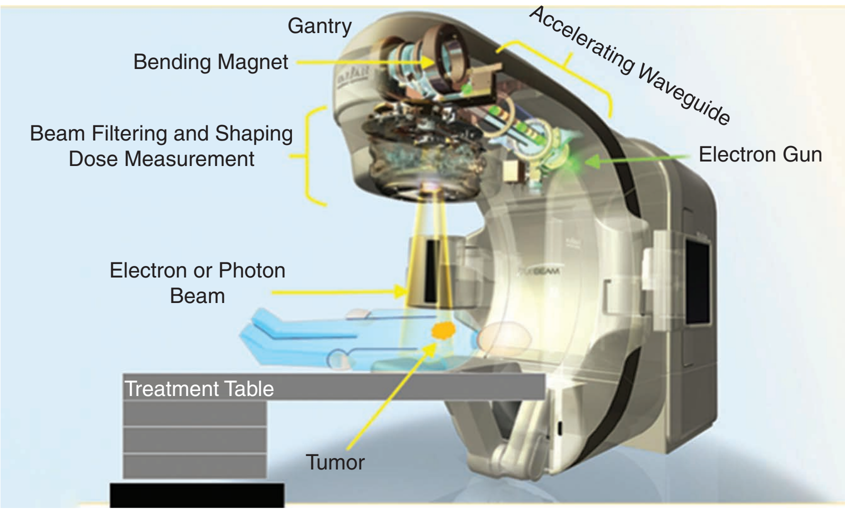
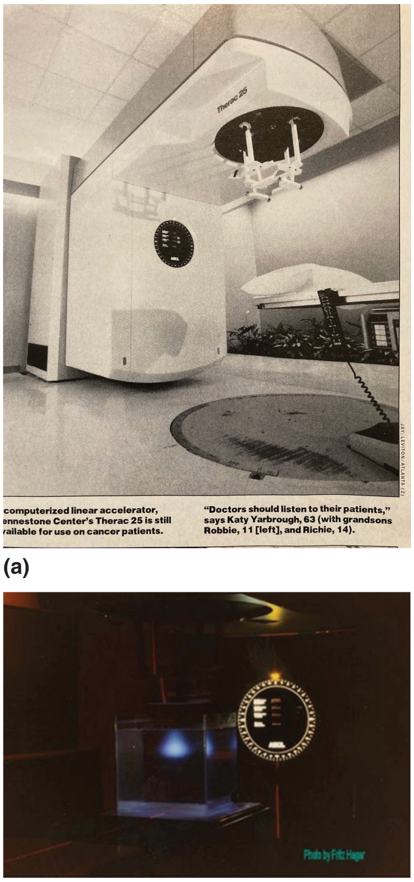


FIGURE 2. A schematic drawing of a medical linac (not the Therac-25). Credits to Varian. Adapted with permission.



(a)

(b)

FIGURE 3. Images from the private collection of Dr. Fritz Hager. (a) A newspaper cut from Plumer[13] showing the "real" Therac-25 (b). The light flash created by the exit electron beam passing through a water phantom simulating a patient during the Malfunction 54 reproduction at ETCC.

collimator-flattening filter combination for X-mode (Figure 4).

Attached to the bottom of the turntable were two separate beam monitors, one for each mode. For safety reasons, each beam monitor consisted of two identical transmission ion chambers. Under the turntable, you had a secondary collimator. Under the second collimator, you had wedges for X-mode or trimmers for E-mode, to optimize the tumor dose distribution. The trimmers were attached to the secondary collimator. You can see the trimmers hanging under the gantry in the photo in Figure 3(a).

PROFESSOR AND STUDENTS: Shall we go back to the day of the first accident?

HAGER: On 21 March 1986, a Friday, I was not in the clinic. They called me and told me that they had a Malfunction 54 error message, and that the patient was hurt and upset.

PROFESSOR AND STUDENTS: Were you familiar with this Malfunction 54 error message?

HAGER: No. We have been treating at the ETCC for two years using a Therac-25, with 60 patients a day, without any serious incidents. The machine used to pause with an error message on the screen saying `MALFUNCTION` followed by a number. We were getting 5–10 malfunction messages every day. If we received a new malfunction, we searched for it in the user manual. But in most cases, it was a false alarm caused by electromagnetic noise. We were used to proceeding further without problems. When we got this new Malfunction 54, we looked in the user manual, but the message was ambiguous; it just said *Dose Error*, which could be

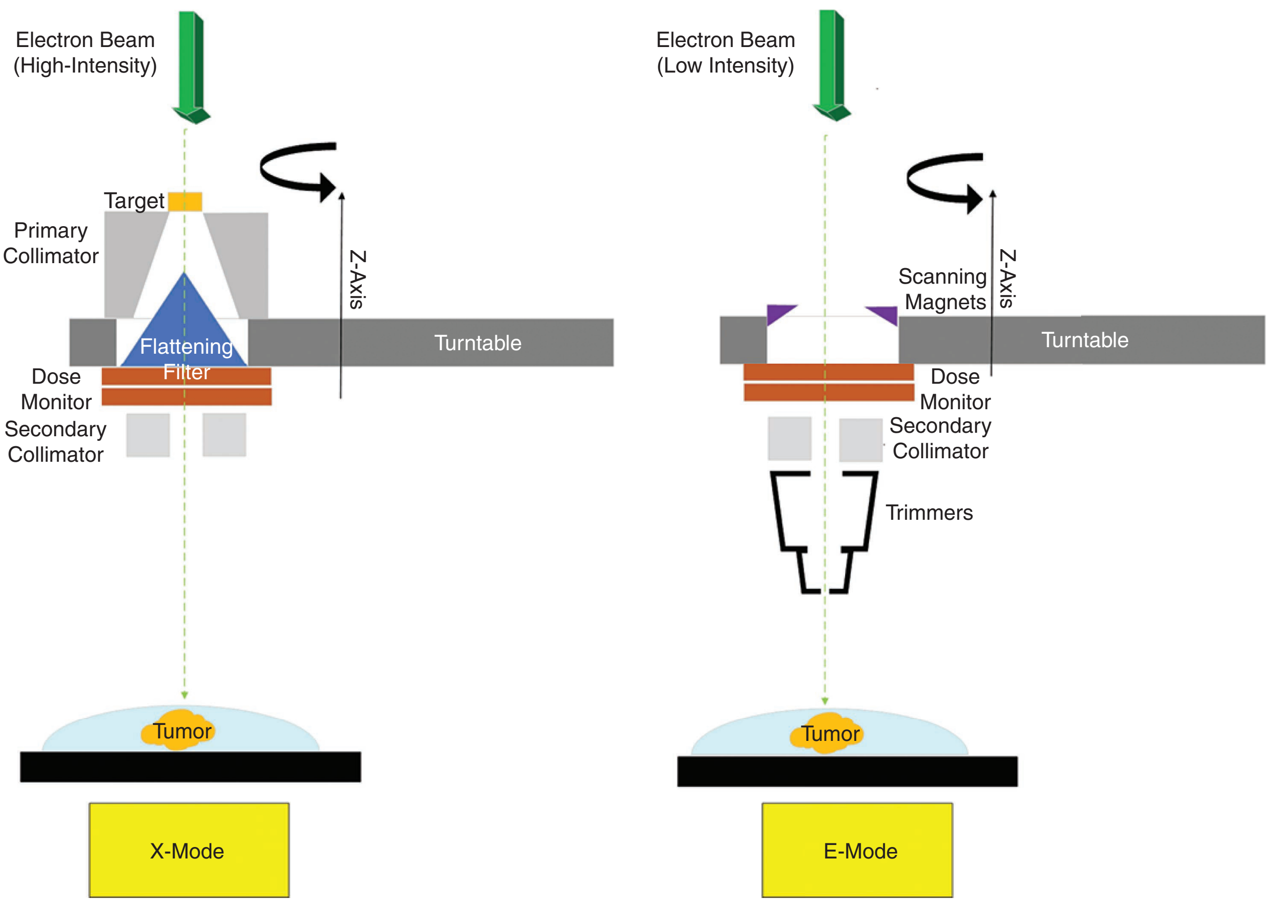


FIGURE 4. The Therac-25 gantry head reconstructed according to Dr. Fritz Hager's recollections.

an underdose or overdose. I think the operator on that day assumed that it was an underdose and resumed the treatment.

PROFESSOR AND STUDENTS: Why didn't you complain to the AECL about the error messages being too cryptic?

HAGER: At that time, we did not feel that the error messages were cryptic, so we did not complain. We did not know better.

PROFESSOR AND STUDENTS: Also, the audio and video communication were nonoperable that day.

HAGER: I was not aware of this problem. I think the operators should have never turned on the machine in this case. A doctor at ETCC examined the patient and found nothing remarkable. The patient went home, and later we lost contact. We notified AECL, and on Monday their engineers came to ETCC. The AECL chief engineer claimed they never had this Malfunction 54 before, and that it was impossible for the machine to generate too high radiation. We hired an engineer to check the ground connections and excluded the possibility of an electric shock. When this seemed fine, we put the machine back in service on 7 April 1986.

PROFESSOR AND STUDENTS: Why did you hurry to put the machine in service?

HAGER: I trusted AECL. At that moment, we did not know that the machine had given an overdose of radiation. Therefore, we also did not notify the FDA (U.S. Food and Drug Administration). Therac-25 was a million-dollar machine and we wanted to use it as much as possible because it had many more treatment capabilities than other machines (Cobalt and Therac-6) we had at the clinic. However, I remember that one oncologist in our center decided not to use the Therac-25 anymore.

PROFESSOR AND STUDENTS: Then a second similar accident happened on 11 April 1986.

HAGER: That day I was in the clinic. The therapist entered my office and said that they got another Malfunction 54. I went immediately to the treatment room. We could still hear a buzzing noise coming from the linac. We did not understand where this noise was coming from. We reported this to AECL. AECL replied that they didn't know why this could happen. The hospital administration instructed us not to talk to journalists or patients. However, I decided to immediately start my investigations.

PROFESSOR AND STUDENTS: Why?

HAGER: You know, I never met the first patient, and I did not even know he was so badly injured by overexposure. In the second case, I was in that room when the patient was getting off the table, I could tell that something serious happened to him, he really looked badly hurt. This burden of responsibility drove me to figure out as soon as possible what happened.

PROFESSOR AND STUDENTS: Were you afraid?

HAGER: Yes. I was afraid I would lose my job. I even thought that if I were at fault, I would have fired myself! The same evening I asked the therapist to tell me what she did exactly to produce a Malfunction 54. I tried to follow her instructions, but couldn't produce the malfunction. We called in another therapist, told her the instructions, and she was able to reproduce the Malfunction 54. The clue was to type fast! With a service ion chamber we could measure a radiation dose of about 9,000 rads, after which the device got saturated. Now I realized that the amount of radiation given to the patient was even more than this, and that the frying eggs noise was created by the saturation of the ion chambers.

When I reproduced myself the Malfunction 54 event and measured the high radiation dose our patients most probably received, I knew that we had killed them. This brought me a feeling of despair. I was sitting in my office that night and thinking: "It is 1986, I am 46 years old, and my career as a medical physicist just ended."

The next day I called AECL and explained to them how to obtain the Malfunction 54. I also called and instructed Tim Still, medical physicist in the Marietta, Georgia hospital who also had a Malfunction 54 event in June 1985. Next, I called all medical physicists who I knew were working on a Therac-25 in the United States and Canada. They were calling me from everywhere saying that they also got the same effect. Not everyone did get the same Malfunction 54 message, but they all could produce the high-dose electron beam. It was now obvious that this was a universal problem, present in all the machines.

PROFESSOR AND STUDENTS: Why did you decide to disobey the instructions of your administration? Did they punish you?

HAGER: I thought about other medical physicists working with a Therac-25. I could imagine how they would feel if they got in the same situation as I did. I could not leave my colleagues clueless as I was. I thought they all must know that they *can* kill a patient. It was further up to them if they still wanted to use the machine after all what happened. Nobody punished or threatened me. AECL lied to me, but not under oath, so this cannot be considered a criminal act.

PROFESSOR AND STUDENTS: The next questions are related to the user interface screen during the Malfunction 54 accident as reproduced in Leveson paper (see Figure 5).[6] The operator made a mistake and typed X instead of E, so this must be the screen for X-mode, right?

HAGER: Yes. The screens for the two modes were almost identical.

MONITOR UNITS (RADS). ACTUAL showed how much radiation in rads was delivered to the patient. There were two values displayed because as I already said, each dose monitor consisted of two identical ion chambers. Only in X-mode, the gantry could rotate. The GANTRY ROTATION (DEG) value indicated the angle in the begin position, which was zero if the beam was directed vertically downwards. The GANTRY ROTATION and other values (COLLIMATOR ROTATION, WEDGE NUMBER, and so on) were manually set by an operator in the treatment room. After that, the set values were transferred to the console in the control room. You could see them on the screen, in the column labeled PRESCRIBED.

AS A RESULT, A HIGH-ENERGY, HIGH CURRENT DENSITY FOCUSED ELECTRON BEAM ENDED UP UNPROTECTED DIRECTLY ON THE PATIENT SKIN, CREATING THE BURN SENSATION AND FATAL TISSUE DAMAGE.

PROFESSOR AND STUDENTS: Were the actual values of GANTRY ROTATION, COLLIMATOR ROTATION, and so on really measured by the software and compared to the prescribed ones, like described in Leveson and Turner?[6]

HAGER: AECL intended to have such a record-and-verify mechanism, but this feature was not fully implemented yet. At that moment, the system only measured the dose given to the patient and compared it to the prescribed dose. The system would stop the treatment when the two were equal. For the rest, the system just read the values set manually by the operator in the treatment room, and *copied* them in the actual value fields on the screen, and displayed the message VERIFIED, but without really executing a verification. This gave a false feeling of safety.

PROFESSOR AND STUDENTS: So the two columns ACTUAL and PRESCRIBED should have exactly the same values.

PROFESSOR AND STUDENTS: Under the OPMODE label, I see "X-RAY." I also see in Leveson and Turner [6] a typical Therac-25 facility drawing, and a box labeled with "14. Turntable position monitor." If these two visual indications of the state of the machine were there, then why did the accidents still happen?

HAGER: No! The mode indicator on the screen and the box in the control room were not there when the accidents happened in 1986. Only in Service mode, we could read the position of the turntable on the display. You could also visually check the position of the turntable using a flashlight, but not during the treatment, of course. All these measures came after the FDA required AECL to design a correction actions plan.

PROFESSOR AND STUDENTS: We conclude (and this is new information) that all of the publicly available drawings show an improved model of Therac-25, manufactured by AECL *after* the accidents.

PROFESSOR AND STUDENTS: Let us walk through the rest of the steps to trigger the MALFUNCTION 54. So, the operator entered all the treatment parameters and the cursor is at the bottom of the screen. A BEAM READY message was expected at the bottom line, but this did not happen. Why?

HAGER: Because the wrong mode set on the console (X), did not match the correct mode set in the treatment room (E). Luckily, there was a hardware interlock that detected this mode inconsistency and was blocking the start of the treatment. You had to go *very fast* (within 8 s) to the top of the screen, using the “Up-arrow” key, change the field BEAM TYPE to the correct E, change the electron beam ENERGY to 10 MeV, and then leave the rest of the fields unchanged by quickly pressing nine times ENTER. As a result, the cursor would arrive at the bottom of the page. This time, the message BEAM READY will get displayed as expected, reassuring the operator that everything is safe, and the Beam can be turned ON. You hit B. Unexpectedly the machine would pause, displaying a very low value in the MONITOR UNITS field (around 6) indicating a low dose, and the message TREAT PAUSED at the bottom line, together with the cryptic MALFUNCTION 54 error message in the field corresponding to REASON. It was possible to resume the treatment for five times by pressing the P key, before the machine would permanently shut down.

PROFESSOR AND STUDENTS: Do you know the state of the machine that triggered the Malfunction-54?

HAGER: Yes. AECL asked me after the accidents to enter the Service mode, repeat the input sequence that led to a Malfunction 54, and print the parameters of the beam just delivered, together with the machine settings. I discovered that the machine was in a combination of modes. The beam delivery part was still set to a 25-MeV X-mode, but the rest of the machine was set to a 10-MeV E-mode, but with no active scanning magnets. A visual examination of the turntable showed that this was also set up for E-mode, with no target in place. As a result, a high-energy, high current density focused electron beam ended up unprotected directly on the patient skin, creating the burn sensation and fatal tissue damage.

PROFESSOR AND STUDENTS: What happened after the accidents?

HAGER: ETCC hired for us an attorney. A few weeks after the accidents, we called Dr. Shalek from Houston to accurately measure the radiation dose using a water tank and a thermo-luminescent dosimeter (TLD). That evening, we had the AECL attorney, our attorney, and AECL engineers together to witness the measurements. We reproduced the Malfunction 54, and a physicist looking at the video monitor in the control room saw a flash of light and the TV monitor turning white from saturation. The next day, we installed a camera in the treatment room and I took a picture of the flash event shown in Figure 3(b). This is how 25,000 rads of 25-MeV electrons delivered in a few seconds looked like.

We also invited Nancy Leveson as a computer science expert, who was able to find the fault which created the Malfunction 54. I had to give a deposition in court. I had the feeling that the lawyers tried to understand the problem, but as they were not physicists, did not know very well which questions to ask. Medical physicists working with Therac-25 participated in a few user meetings with AECL, FDA, and hospital administrators. During these meetings, I think the physics community came up with more solutions to make the system safe than others did. There were no software people in these meetings as far as I can remember. Because AECL was a crown corporation, the case was settled in court. AECL had a liability sum of US$10 million that was divided among the injured patients and their families.

I found that AECL was slow in making changes. I was disappointed that their first corrective measure was to tape the Up-arrow key to disable editing. I personally refused to use the Therac-25, even after AECL operated all these changes. I suspected it was too much

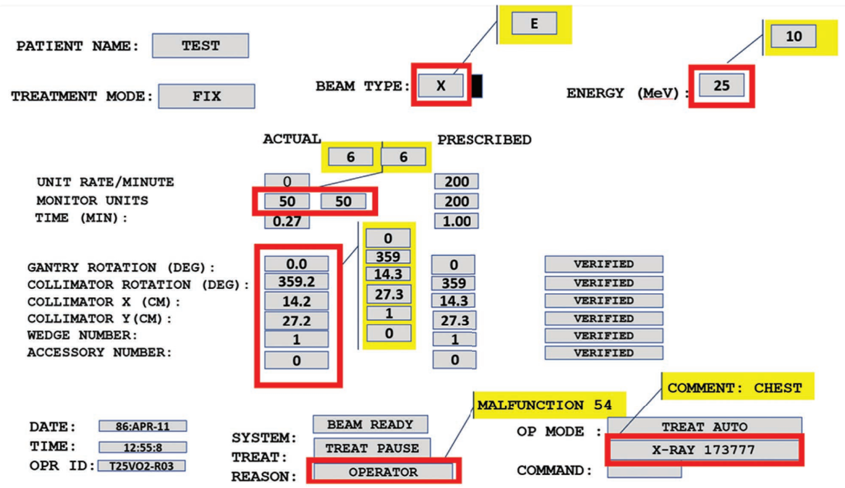


FIGURE 5. The Therac-25 operator interface screen after the system paused with a Malfunction 54 error message. We changed the values in a red frame from Leveson and Turner[6] to the values in a yellow frame based on Dr. Fritz Hager’s recollection.

wrong there in their design. AECL took our Therac-25 back and all its documentation, and paid us US$1 million to buy a new machine. I think that this settlement indicates that AECL took responsibility for the accident.

I also think they were pretty arrogant. Let me ask you. You write software. How many times it works from the first time? Almost never. You need to test it. And we, the radiation therapists, expect from the manufacturer to provide us with a safe machine that never allows a lethal dose of radiation to reach the patient. We do not have the capability to test and find bugs. So we trust you.

PROFESSOR AND STUDENTS: Can you describe the emotional impact the accidents had on you?

HAGER: First I felt very upset and was afraid that it was all my fault, and that I could have prevented all this. Discovering that it was not my fault was the first step to recovery. The second thing that helped was a psychologist who came in and talked to the whole staff. We met in the Therac-25 room, which was very emotional because this was the very place where our patients got their injuries. We talked about the patients and the experience we all had; it was a good time of sharing and opening. However, I will always regret that we had two incidents instead of one. Maybe if I had questioned the therapist more thoroughly after the first incident, a clue might have come up and we could have prevented the second incident. I also wish I had access to our first victim. You know, events like these are getting embedded in your memory and you don't forget much; even the little details you still remember. It was so tragic all what happened.

PROFESSOR AND STUDENTS: Do you have peace now?

HAGER: Yes. I think I did everything I could. I discovered the mechanism that caused the accidents and probably saved other lives. When I understood that the overdoses were caused by a very rare, difficult-to-detect software fault and this happened in many clinics, without any warning from the machine, I decided to stay in the RT field. I worked for 46 years in RT, and I never witnessed something like the Therac-25 accidents.

PROFESSOR AND STUDENTS: Except for improved safety regulations, what was in your opinion the impact of the Therac-25 accidents on the RT field?

HAGER: The effect of the accidents was a damaged trust in computer-controlled RT. For five to seven years after the accidents, we did not have linacs controlled by software. Now things are much safer. It seems manufacturers had to learn to make safe machines based on these accidents. Ironically, recently I read about a new RT technique, called FLASH radiotherapy.[14] In fact, what they do is to use a 10-MeV electron beam without a target or flattening filter, reproducing in fact the hazard that caused the Malfunction 54 radiation overdose. So, you see how things go.

PROFESSOR AND STUDENTS: One last question. Why did you agree to speak to us?

HAGER: Therac-25 accidents are unique, like no other accidents in RT. It is important for people to know how they occurred. In particular, it is important that future software developers keep these things in mind. People should know that Therac-25 was a great machine, but in RT, when things go wrong, they can go very wrong. I want to inform even now as many people as possible. I believe that events such as the Therac-25 should be freely and widely reported so that others will not make the same mistakes. It is much better to learn lessons in the classroom rather than when peoples' lives are at risk.

PROFESSOR AND STUDENTS: Thank you very much, Fritz for sharing with us your memories and insights.

HAGER: Thank you for bringing this accident forward to students concerned with computer safety. I hope they will do a good job!

REFLECTION

Therac-25 accidents were extensively investigated, and many lessons have already been learned.[6,15,16,17,18] Since then, our understanding and capabilities in software engineering have improved dramatically. What else can be done to prevent similar accidents from happening, especially now, at the dawn of another revolutionary technology such as artificial intelligence? In our opinion, software engineers interested in building mission-critical systems should gain solid knowledge on the following topics.

1. *Quality Assurance (QA)*: Therac-25 accidents showed how suboptimal QA can turn software systems into lethal weapons. The problem is that at this moment, by far not all computing curricula feature a dedicated software testing course. Therefore, we urge all software engineers to learn how to build *test* cases to show that their software behaves as expected. The next level, a more laborious step, is to learn how

to use formal methods to *prove* that their software is correct. Unfortunately, a good-for-all QA strategy does not exist. Instead, sound QA guidelines tailored for a specific domain, like RT need to be formulated.[19] This can be achieved only through a close collaboration between software manufacturers and RT domain experts (medical physicists, oncologists and therapists) during the whole development process. Even after this, when incidents happen in the field, software engineers should take part in investigations in order to improve their practices. Talking to witnesses will make the learning effect even stronger.

2. *Safety Science*: Therac-25 accidents did not happen because of one faulty component; they happened because a faulty interaction between components. For example, the software changed the treatment mode from X to E *faster than* the hardware could, or a *feedback* channel from the hardware to software *was missing*. This means that ideally, a safe design should start by identifying what can go wrong with the *entire* system including its hardware and users. Although strict safety standards regulate nowadays the engineering of mission-critical software systems, none of them specifies which method to use for this so-called hazard analysis. We recommend familiarizing software engineers with the relatively new systems theory-based accident modeling and processes (STAMP) framework, designed by the same Leveson, 20 years after the Therac-25 accidents.[20] An advantage of STAMP is that it can identify *interaction-type* of hazards. Moreover, STAMP can be conducted by an outsider such as a computer professional together with domain experts, even when the system is not yet implemented. Formal methods can help here again to detect behavior that is not specified. New safety-related requirements, design constraints and test cases will emerge from this hazard analysis, which will lead to a more robust system and hopefully less accidents.

3. *Human Factors*: The Malfunction 54 accidents have shown that users are not machines; they operate software with the best intentions, and they trust it, but can be tired, distracted or stressed, too hesitant to admit that they do not understand error messages, or too confident they can neglect certain alerts. These are dangerous conditions that a programmer must anticipate and mitigate, for example with informative and unambiguous error messages that are also not too frequent, to avoid alarm fatigue. Also, we recommend that after a linac software paused with a severe error message, a very rigorous procedure should be in place to decide whether to resume the treatment or not. Although these recommendations were already formulated by the Therac-25 investigators, we believe they are still actual today. The reason is a survey we recently conducted among the RT technicians worldwide, which revealed, for example, that up to 25% of the respondents categorized the error messages they received as unclear.

We reported on a series of conversations we had with an eyewitness of two Therac-25 accidents. The results include new visual material (a photo of the real machine and of the huge light flash created by the faulty machine in a water phantom, a more accurate reconstruction of the linac's gantry and its user interface) and new details related to the aftermath of the accidents, together with an emotional second-victim testimony and some lessons learned. In the future, we will investigate how to apply STAMP to prevent accidents in RT. We will also keep on encouraging witnesses of other computer-related accidents to open a dialog with new generations, to share their memories, and reflect together on how to build a safer, software-intensive society. ■

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